HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F79SP

STOCHASTIC PROCESSES

IN-PERSON EXAMINATION

December Examination and Assessment Diet 2023

Duration: 2 hours

Total marks available: 80 marks

Answer all 5 questions

Attached: Formula sheet and Standard Normal Distribution table included
Supplied: No additional material supplied
Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

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Some Absolutely Continuous Distributions

Name	Notation	p.d.f.	range	m.g.f.	mean	variance
Uniform	$U(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{b\theta}-e^{a\theta}}{(b-a)\theta}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu\theta+\frac{1}{2}\sigma^2\theta^2}$	μ	σ^2
Gamma	$\Gamma(t,\lambda)$	$\frac{\lambda^t}{\Gamma(t)}x^{t-1}e^{-\lambda x}$	$0<x<\infty$, $t,\lambda>0$	$\left(\frac{\lambda}{\lambda-\theta}\right)^t$	$\frac{t}{\lambda}$	$\frac{t}{\lambda^2}$
Exponential	$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-\theta}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_n	$\frac{1}{2^{n/2}\Gamma(n/2)}x^{n/2-1}e^{-x/2}$	$0<x<\infty$, $n>0$ integer	$(1-2\theta)^{-n/2}$	n	$2n$
Beta	$\beta(\alpha_1,\alpha_2)$	$\frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)}x^{\alpha_1-1}(1-x)^{\alpha_2-1}$	$0<x<1$, $\alpha_1,\alpha_2>0$	not useful	$\frac{\alpha_1}{\alpha_1+\alpha_2}$	$\frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)}$
Cauchy	–	$\frac{\lambda}{\pi(\lambda^2+(x-\mu)^2)}$	$-\infty<x<\infty$, $\lambda>0$	doesn't exist but char. fn. is $e^{i\mu\theta-\lambda\lvert\theta\rvert}$	no moments exist	
Multivariate Normal	$N(\mu,\Sigma)$	$\frac{\exp\left\{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right\}}{(2\pi)^{d/2}\det(\Sigma)^{1/2}}$	$x\in\mathbb{R}^d$	$e^{\mu^T\theta+\frac{1}{2}\theta^T\Sigma\theta}$	μ	covariance matrix Σ

Some Discrete Distributions

(All distributions below have integer-valued ranges)

Name	Notation	prob. fn.	range	m.g.f.	mean	variance
Binomial	$\text{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$0\le x\le n$ $0<p<1$	$[(1-p)+pe^\theta]^n$	np	$np(1-p)$
Poisson	$\text{Pois}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$0\le x<\infty$ $\lambda>0$ real	$e^{\lambda(e^\theta-1)}$	λ	λ
Hyper-geometric	$\text{Hgeo}(n,a,b)$	$\frac{\binom{a}{x}\binom{b}{n-x}}{\binom{a+b}{n}}$	$0\le x\le n$ $a,b>0$ integers	not useful	$\frac{na}{a+b}$	$\frac{nab(a+b-n)}{(a+b)^2(a+b-1)}$
Negative binomial	$\text{Nbin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$k\le x<\infty$, $0<p<1$	$p^k[e^{-\theta}-(1-p)]^{-k}$	k/p	$k(1-p)/p^2$
Geometric	$\text{Geo}(p)$	$p(1-p)^{x-1}$	$1\le x<\infty$ $0<p<1$	$p[e^{-\theta}-(1-p)]^{-1}$	$1/p$	$(1-p)/p^2$

Multinomial: multivariate extension of binomial

Prob. fn. is given by

$$f(x_1,x_2,\ldots,x_k)=\frac{n!}{x_!x_2!\ldots x_k!x_{k+1}!}p_1^{x_1}p_2^{x_2}\ldots p_k^{x_k}p_{k+1}^{x_{k+1}},\quad 0\le x_i\le n,\ \ 0<p_i<1\ \ \forall i=1,2,\ldots,n+1$$

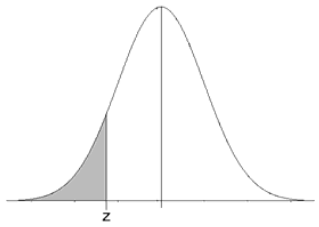
where $p_1+p_2+\cdots+p_{k+1}=1$ and $x_1+x_2+\cdots+x_{k+1}=n$.

The m.g.f. is given by

$$[p_1e^{\theta_1}+p_2e^{\theta_2}+\cdots+p_ke^{\theta_k}+p_{k+1}]^n.$$

Moments: $E(X_i)=np_i$, $\text{Var}(X_i)=np_i(1-p_i)$, $\text{cov}(X_i,X_j)=-np_ip_j$, $i,j=1,2,\ldots,n+1$.

Standard Normal Cumulative Probability Table



Cumulative probabilities for NEGATIVE z-values are shown in the following table:

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

1. Let $S_n = \sum_{i=1}^{n} X_i$ be a random walk with initial state $S_0 = 0$, where $X_1, X_2, ...$ are independent and identically distributed random variables with

$$\mathbf{P}(X_i = 1) = 1/2, \quad \mathbf{P}(X_i = 0) = 1/3, \quad \mathbf{P}(X_i = -1) = 1/6,$$

(i.e. the walk jumps up with probability $p = 1/2$, has no jump with probability $r = 1/3$, and jumps down with probability $q = 1/6$).

(a) Calculate exactly $\mathbf{P}(S_2 \geq 0)$.

[3 marks]

(b) Find an approximate value of $\mathbf{P}(S_{100} \geq 40)$.

[3 marks]

(c) Determine: $\mathbf{P}$(there exists some $n > 0$ such that $S_n \leq -4$).

[3 marks]

(d) Determine: $\mathbf{P}$(there exists some $n > 0$ such that $S_n \geq 15$).

[2 marks]

[Total: 11 marks]

2. A Markov chain $\{X_n\}_{n\geq 0}$ takes values in the set $\{1,2,3,4,5\}$ and has transition matrix

$$\begin{pmatrix} 1/4 & 0 & 0 & 0 & 3/4 \\ 1/2 & 1/8 & 1/8 & 1/4 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1/6 & 1/6 & 1/3 & 1/6 & 1/6 \\ 2/3 & 0 & 0 & 0 & 1/3 \end{pmatrix}$$

(a) Identify the open and the closed classes of this Markov chain. Provide your reasoning.

[3 marks]

(b) Determine the limiting distribution of the Markov chain if it starts from the initial state $X_0 = 5$.

[5 marks]

(c) For $1 \leq i \leq 5$, let y_i denote the probability of absorption in state 3 if the Markov chain starts from state $X_0 = i$. Determine the values of y_i for $1 \leq i \leq 5$.

[5 marks]

(d) Determine the probability that the Markov chain hits the set $\{1,5\}$ if it starts from state $X_0 = 4$.

[3 marks]

(e) Determine the limiting distribution of this Markov chain if it starts from $X_0 = 4$.

[3 marks]

(f) Determine the family of all stationary distributions for this Markov chain.

[3 marks]

[Total: 22 marks]

3. Consider a Galton-Watson branching process $\{X_n\}_{n\geq 0}$ in which the probability p_k that an individual has exactly k offspring is given by

$$p_0 = x/3, \quad p_1 = 2/3, \quad p_2 = (1-x)/3, \quad p_k = 0 \ \text{ for } \ k \geq 3$$

where $x \in [0,1]$ is a fixed parameter. Let $T = \min\{n \geq 0 \ : \ X_n = 0\}$ denote the extinction time of the branching process $\{X_n\}$.

(a) Assume the initial state is $X_0 = 1$. Determine the values of the parameter x for which the extinction probability $\mathbf{P}(T < \infty) = 1$.

[2 marks]

(b) Now suppose that the initial state is $X_0 = 4$. In this case, for which values of x do you obtain $\mathbf{P}(T < \infty) = 1$?

[2 marks]

(c) Assume (again) that the initial state is $X_0 = 1$ and suppose that the parameter $x = 1/3$.

i. Find $\mathbf{P}(T = 3)$.

[6 marks]

ii. Using the probability generating function, state (without proof) the general formula for obtaining the extinction probability $\varepsilon = \mathbf{P}(T < \infty)$ of a branching process. Use this formula to determine the value of ε when $X_0 = 1$ and $x = 1/3$.

[5 marks]

[Total: 15 marks]

4. (a) Assume that random variables $\xi_1, \xi_2, \ldots$ are i.i.d., with a common Exponential distribution with parameter $\lambda > 0$. Let $S_0 = 0$ and $S_n = \xi_1 + \ldots + \xi_n$, for $n = 1, 2, \ldots$. For $t > 0$, let $N(t) = \max\{n \geq 0 \ : \ S_n \leq t\}$ and $\psi(t) = S_{N(t)+1} - t$.

Provide complete proofs of the following three facts:
(i) $N(t)$ has a Poisson distribution with parameter λt;
(ii) $\psi(t)$ has an Exponential distribution with parameter λ;
(iii) $N(t)$ and $\psi(t)$ are mutually independent random variables.

[9 marks]

(b) In the city of Edingow the council has decided to build a 24-hour car park with infinite capacity. The car park opens on its first day of business at noon and cars begin to arrive to park according to a Poisson process with rate $\lambda = 30$ cars per hour.

Independently of everything else, each car driver buys a one-hour ticket (short-stay) with probability $p = 2/3$ or a three-hour ticket with probability $1 - p = 1/3$.

i. Determine the distribution of the number of arrivals at the car park by 4 pm on the first day of business.

[2 marks]

ii. Determine the distribution of the number of short-stay tickets that have been sold by 4 pm on the first day.

[2 marks]

iii. Determine the conditional expected number of arrivals before 3 pm on the first day, given that 100 cars have arrived by 4 pm. (Give reasons for your answer.)

[5 marks]

Hint: You may use the Superposition and Splitting theorems.

[Total: 18 marks]

5. A continuous-time, homogeneous Markov process $\{X(t)\}_{t\geq 0}$ takes values in the set $\{1, 2, 3, 4, 5\}$ and has a transition rate matrix Q of the form

$$\begin{pmatrix} -8 & 0 & 0 & 2 & 6 \\ 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & -4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & -5 \end{pmatrix}$$

(a) Classify the states of the Markov process $\{X(t)\}_{t\geq 0}$ and draw the communication diagram for this process.

[4 marks]

(b) Find all stationary distributions for this Markov process.

[5 marks]

(c) Find the transition matrix P^* of the embedded (jumping) Markov chain.

[2 marks]

(d) **Without calculating the stationary distribution of the embedded Markov chain**, determine whether the families of the stationary distributions of the Markov process $\{X(t)\}_{t\geq 0}$ and of the embedded Markov chain coincide or not? Give reasons for your answers.

[3 marks]

[Total: 14 marks]

END OF PAPER